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# Newton-Raphson Method

def f(x):
    return x**3 - x - 2

def df(x):
    return 3*x**2 - 1

x0 = float(input("Enter initial guess: "))
tolerance = float(input("Enter permissible error: "))

iteration = 0

while True:
    fx = f(x0)
    dfx = df(x0)

    x1 = x0 - (fx / dfx)

    error = abs(x1 - x0)

    iteration += 1
    print(f"Iteration {iteration}: x = {x1}, Error = {error}")

    if error < tolerance:
        break
    else:
        x0 = x1

print("\nRoot is:", x1)

===== RESTART: D:\Experiment 2\Newton Raphson.py
=====
Enter initial guess: 1
Enter permissible error: 2
Iteration 1: x = 2.0, Error = 1.0

Root is: 2.0

```